

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I M Nishanth (192511154) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Nishanth

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25 School of

Electronics → /26

School of Mechanical Engineering → /27

2. As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

3. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.

3. Calculate the **number of usable hosts** in each subnet.

4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.

2. Expand the compressed address back into its **full hexadecimal representation**.

3. Identify the **network prefix** using the /64 prefix length.

4. Calculate the **total number of host addresses** supported within the subnet.

5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.

3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

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ASSESSMENT TOOL 2 – CSA0706 COMPUTER NETWORKS (CO2)

Annexure A – Scenario-Based: Worked Solutions

Question 1: Smart Campus Subnet Design (192.168.100.0/24)

The /24 block 192.168.100.0/24 must be divided into three unequal subnets using VLSM (Variable Length Subnet Masking), allocating the largest subnet first so that address blocks align correctly.

1 & 2 & 3. Network Address, Broadcast Address, Host Range and Usable Hosts

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.127	192.168.100.1 – 126	126
School of Electronics	/26	192.168.100.128	192.168.100.191	192.168.100.129 – 190	62
School of Mechanical Engg.	/27	192.168.100.192	192.168.100.223	192.168.100.193 – 222	30

Calculation basis: /25 gives $2^7 - 2 = 126$ hosts (128-address block); /26 gives $2^6 - 2 = 62$ hosts (64-address block); /27 gives $2^5 - 2 = 30$ hosts (32-address block).

4. Remaining Unused IP Address Range

Total addresses used so far = 128 (Computing) + 64 (Electronics) + 32 (Mechanical) = 224 addresses, out of 256 available in the /24 block (192.168.100.0 – 192.168.100.255).

Unused/reserved range = 192.168.100.224 – 192.168.100.255 (32 addresses, equivalent to a /27 block)

This remaining block can be reserved for future departments or expansion, such as a new lab or administrative unit.

Question 2: Multinational Company Subnetting (10.10.0.0/16)

The private block 10.10.0.0/16 (65,536 addresses) is divided using VLSM into four subnets, allocated sequentially starting from the largest (/24) to the smallest (/27).

1, 2 & 3. Network Address, Broadcast Address, Host Range and Usable Hosts

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	/25	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	/26	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	/27	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	30

Calculation basis: /24 gives $2^8 - 2 = 254$ hosts; /25 gives $2^7 - 2 = 126$ hosts; /26 gives $2^6 - 2 = 62$ hosts; /27 gives $2^5 - 2 = 30$ hosts.

4. Unused IP Address Range Remaining

The four departments together occupy addresses from 10.10.0.0 through 10.10.1.223, a total of $256 + 224 = 480$ addresses used out of the full 65,536 addresses available in the /16 block.

Unused range = 10.10.1.224 – 10.10.255.255

This very large remaining address space (well over 65,000 addresses) is available for future departments, new sites, or additional growth within the organization.

Question 3: IPv4-to-IPv6 Migration for Smart Hospital

Given IPv6 network address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed Notation

Compressed form = 2001:db8:abcd::

Rule applied: leading zeros in each 16-bit group are dropped, and the longest continuous run of all-zero groups (four consecutive zero groups here) is replaced once with '::'.

2. Expanded (Full) Notation

Expanded form = 2001:0db8:abcd:0000:0000:0000:0000

3. Network Prefix (/64)

Network prefix = 2001:0db8:abcd:0000 /64 (i.e., 2001:db8:abcd::/64)

This means the first 64 bits (the first four hextets) identify the hospital's network, while the remaining 64 bits are available for host/interface identifiers.

4 & 5. Total Number of Host Addresses within the /64 Subnet

Host bits available = $128 - 64 = 64$ bits

Total possible host addresses = $2^{64} \approx 1.844674 \times 10^{19}$ addresses

This enormous address space is more than sufficient to uniquely address every medical device, IoT sensor, and connected equipment throughout the hospital, with massive room for future expansion – a key reason IPv6 is well suited for large-scale IoT deployments such as smart hospitals.

Question 4: Routing Table Lookup at ISP Core Router

Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24. Destination IP of arriving packet: 192.168.2.45.

1. Destination Subnet

192.168.2.45 lies within the range 192.168.2.0 – 192.168.2.255, so the packet belongs to the subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router performs a longest-prefix match against its routing table. Comparing the destination address to each entry, only 192.168.2.0/24 matches the network portion of 192.168.2.45. Hence, the matching entry is 192.168.2.0/24.

3. Next-Hop Network / Interface

The router forwards the packet out of the interface directly connected to (or configured as the next hop for) the 192.168.2.0/24 network – i.e., the router's interface serving that customer network/LAN segment.

4. Default Route (No Match Case)

If the destination address did not match any of the three specific routing table entries, the router would forward the packet using its default route, typically represented as 0.0.0.0/0, which directs unmatched traffic toward the ISP's upstream gateway/next-hop router for onward delivery to the wider Internet.

Question 5: Layer-3 Routed Office Network (Administration & Finance LANs)

Administration LAN = 192.168.10.0/26, Finance LAN = 192.168.10.64/26, connected via a Layer-3 router.

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Subnet	Network Address	Broadcast Address	Valid Host Range
Administration	192.168.10.0/26	192.168.10.0	192.168.10.63	192.168.10.1 – .62
Finance	192.168.10.64/26	192.168.10.64	192.168.10.127	192.168.10.65 – .126

Each /26 subnet provides $2^6 - 2 = 62$ usable host addresses.

3. Suggested IP Addresses for Three Computers in Each LAN

Administration LAN (192.168.10.1 – .62 usable, with .1 reserved for the gateway): PC1 = 192.168.10.2, PC2 = 192.168.10.3, PC3 = 192.168.10.4

Finance LAN (192.168.10.65 – .126 usable, with .65 reserved for the gateway): PC1 = 192.168.10.66, PC2 = 192.168.10.67, PC3 = 192.168.10.68

4. Default Gateway for Each LAN

Hosts in the Administration LAN should be configured to use default gateway 192.168.10.1 (the router's Administration-facing interface).

Hosts in the Finance LAN should be configured to use default gateway 192.168.10.65 (the router's Finance-facing interface).

Since both LANs are connected through the same Layer-3 router, the router can route traffic between the Administration and Finance departments (inter-VLAN/inter-subnet routing) as well as forward traffic to external networks.

